Multimedia Signal Processing - 2nd Module

February,17 2022

1. **Question (8 pts)** Consider the problem of developing a lossless coding solution for words of 2 letters considering the limited alphabet $\{A, B, C, D\}$. Each word can be seen as a pair (x, y) obtained by sampling one symbol x from the source X and one symbol y from the source Y. After analyzing your dictionary of 2-letter words, you derive the following p.m.f.

	$x = A$	$x = B$	$x = C$	$x = D$
$y = A$	1/8	1/16	0	1/16
$y = B$	0	0	1/8	0
$y = C$	1/8	2/8	0	1/8
$y = D$	0	1/16	0	1/16

   (a) Determine the p.m.f that characterizes the single source X.

   (b) Compute the entropy of X (i.e., $H(X)$)

   (c) Design the Huffman code that encodes single letters from X and compute the average code length.

   (d) Compute the joint entropy of X and Y (i.e., $H(X, Y)$).

   (e) Design a prefix-code to encode 2-letter words.

   (f) Measure the average code length of your prefix-code and report how far you are from the theoretical optimal bound.

   (g) Given your prefix-code, which is the extension code for the word AC?

   (h) Given your prefix-code, which is the extension code for the word AD?

2. **Question (4 pts)** With the help of a block diagram, describe the typical hybrid DPCM/DCT video coding scheme. Explain each block, and clearly distinguish the encoder from the decoder.

3. **Question (3 pts)** Consider two DMS sources X and Y. Define the concept of marginal (i.e., $H(X)$), joint (i.e., $H(X, Y)$) and conditional (i.e., $H(X|Y)$) entropy. With the help of a Venn Diagram, report the relationships among these quantities.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Mutual information between two discrete sources X and Y cannot be negative.
   - [T] [F] A symbol a with probability $p(a) = 0.5$ has information $i(a) = 1$ bit.
   - [T] [F] Given a discrete source, the average code length of the Shannon code can be smaller than the average code length of the Huffman code.
   - [T] [F] According to rate-distortion theory, it is not possible to encode a discrete source with a distortion greater than the source variance.
   - [T] [F] If lossy coding is applied to a source (e.g., quantization), it is useless to apply a further step of lossless coding.
   - [T] [F] Consider a vocoder exploiting LPC. It is not safe to quantize LPC coefficient directly, as it can cause instability.
   - [T] [F] Both DCT and KLT are signal-dependent transforms.
   - [T] [F] In the JPEG standard, the luminance component is often sub-sampled as the chrominance is more important for the human visual system.

5. **Question (Matlab - 10pts)**

   Compare different quantizers applied to a mono-dimensional signal.

   Signal generation

   (a) Generate an auto-regressive process of order one (i.e., AR(1)) of 500 samples. Consider the model $x(n) = \rho * x(n-1) + w(n)$, where $\rho = 0.95$ and $w(n)$ is extracted from a Gaussian distribution with zero mean and standard deviation equal to 2.

   (b) Clip the values of this signal in the range $[-15, 15]$. This is your input signal `x`.

   Quantizers

   (a) Quantize `x` obtaining `y_1` using a uniform scalar midrise quantizer with quantization step $\Delta = 1$.

   (b) Quantize `x` obtaining `y_2` using a uniform scalar midthread quantizer with quantization step $\Delta = 2$.

   (c) Quantize `x` obtaining `y_3` using a quantizer with the following quantization rule

$$y = q(x) = \begin{cases} -10 & \text{if } x \leq -5, \\ 0 & \text{if } -5 < x \leq 5, \\ 10 & \text{if } x > 5. \end{cases}$$

   Comparison

   (a) Compare the three quantizers by plotting on a graph the rate-distortion working point of each one of them. Use entropy as rate, and MSE as distortion.